

Law of the Unique Price.

Assume that the payoffs of two static portfolios A and B are equal, i.e.,

$$V_A(T) = V_B(T)$$

Claim. $V_A(0) = V_B(0)$

Proof. Assume, to the contrary, that

$$V_A(0) \neq V_B(0). \quad \leftarrow$$

Without loss of generality,

$$\underbrace{V_A(0)}_{\text{relatively cheap}} < \underbrace{V_B(0)}_{\text{relatively expensive}}$$

Diagnosis.

Propose an arbitrage portfolio:

- Long Portfolio A
 - Short Portfolio B
- } Total Portfolio

Verify that this is, indeed, an arbitrage portfolio.

$$\bullet \text{ Initial Cost (Total Portfolio)} = V_A(0) - V_B(0)$$

< 0
Initial Inflow
of Money

$$\bullet \text{ Payoff (Total Portfolio)} = V_A(T) - V_B(T) = 0$$

$$\begin{aligned} \text{Profit} &= \text{Payoff} - FV_{0,T}(\text{Initial Cost}) \\ &= 0 - FV_{0,T}(V_A(0) - V_B(0)) > 0 \end{aligned}$$

Indeed, this is
an ARBITRAGE
PORTFOLIO!

$\Rightarrow \Leftarrow$



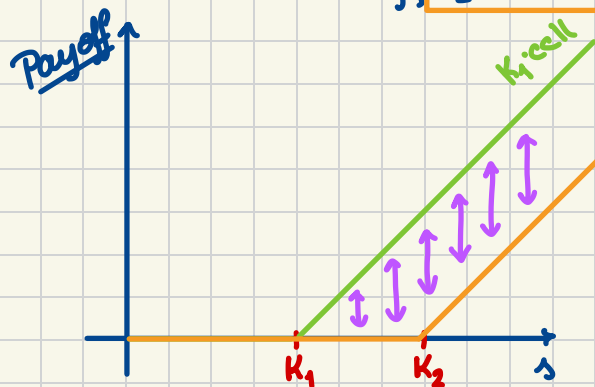
Corollary. If $V_A(T) \geq V_B(T)$, then

$$V_A(0) \geq V_B(0)$$

Example. $K_1 < K_2$

A: one long, K_1 -strike call
B: one long, K_2 -strike call

European and
w/ the same
underlying asset
& exercise date



The payoff of the
 K_1 -strike call
dominates
the payoff of the
 K_2 -strike call.

$\Rightarrow$ The K_1 -call costs @ least as much as
the K_2 -call.

In Math:

$$K_1 < K_2 \Rightarrow V_C(0, K_1) \geq V_C(0, K_2)$$

As a function of the strike price,
call prices are DECREASING.

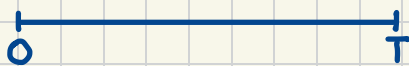


Replicating Portfolio.

Def'n. Consider a European-style derivative security.
A static portfolio w/ the same payoff as that of the derivative security is called its replicating portfolio.

Note: Because we assume no arbitrage, the initial price of the derivative security is equal to the initial price of its replicating portfolio.

Example. Consider a forward contract on a non-dividend-paying STOCK/INDEX.



Forward Contract: $S(T) - F$

Replicating Portfolio: $\left\{ \begin{array}{l} \bullet \text{ long one share of stock} \\ \bullet \text{ issue a bond w/ redemption amt } F \text{ and maturity } T \end{array} \right.$

Payoff (Portfolio) = $S(T) - F$

NO ARBITRAGE!

$\Rightarrow$ The forward contract and its replicating portfolio must have the same initial cost, i.e.,

$$0 = S(0) - PV_{0,T}(F)$$

$$\Rightarrow PV_{0,T}(F) = S(0) \Rightarrow F = FV_{0,T}(S(0)) = S(0)e^{rT}$$

Example.

$$\text{Payoff} = \max \left(\frac{S(T)}{S(0)}, (1+g)^T \right)$$
$$= \frac{1}{S(0)} \max (S(T), S(0)(1+g)^T)$$

$$V_p(T) = (K - S(T))_+ = \max (K - S(T), 0)$$

$$\frac{1}{S(0)} \left(\underbrace{S(T)}_{\substack{\uparrow \\ \text{long stock}}} + \underbrace{\max (0, S(0)(1+g)^T - S(T))}_{\substack{\uparrow \\ \text{put payoff}}} \right)$$